

DriveLearn India

Multi-Portal System Architecture & Feature Specification Dossier

Compliance with Central Motor Vehicles Rules (CMVR 1989) • Maharashtra RTO Digital Network

Executive Summary: DriveLearn India is an end-to-end multi-portal motor driving academy ecosystem built for Maharashtra RTO jurisdictions (Pune MH-12, Mumbai MH-02, Nagpur MH-31, Nashik MH-15, Thane MH-04). The architecture unifies five dedicated role-based portals, resolving real-world logistical pain points such as dual-control car telemetry, monsoon weather delays, zero-phone-tag transit pickups, and official Form 5 certification.

Platform Ecosystem & Role Interconnection Architecture

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DRIVELEARN STATE PLATFORM CLOUD |
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| v v [ 1. PUBLIC WEBSITE ] [ 5. SUPER ADMIN ] * School Search & GPS * RTO Compliance Desk * Subsidized
Rs.999 Offer * School Licensing Verification * Parivahan RTO Mock Test * Platform Wallet Subsidies |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Instant Student Onboard | v v [ 2. LEARNER HUB ] [ 3. INSTRUCTOR COCKPIT
] * Real-Time Dashboard * 45-Min In-Cockpit Console * Explore Courses & Schools * Live Dual-Control
Drills * Interactive Slot Booking * Monsoon Rain Delay Broadcaster * Safety Shield (WhatsApp SOS) *
Student Skill Sign-Offs * CMVR Form 5 Passing Cert * Transit Landmark Pickups ^ ^ | Fleet & Roster |
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[ 4. SCHOOL OWNER ] -----+ * Fleet Maintenance & RC Ledger * Fuel / CNG
Operational Audit * Instructor Payroll & Batches * Form 5 Generation & Revenue
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1. Public Website & Discovery Portal

Operational User Journey Flow:

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[Prospective Student] | v +-----+ +-----+ +-----+ +-----+ +-----+ |
Home Banner | ---> | Find Schools | ---> | School Profile | ---> | Reserve Package | ---> [Login /
Student Hub] | (Launch Offer)| | (GPS & RTO MH) | | (Fleet & Revs) | | (Apply Rs.15 Bonus)|
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Module / Feature	Route / Component	One-Line Technical & Operational Purpose
Landing Page	/ (Home.jsx)	Showcases Maharashtra subsidized Rs.999 launch, pass rates, and safety guarantees.
Find Driving Schools	/find-school	Geo-filtered directory searching RTO-certified schools with city and GPS radius filters.
School Dossier Profile	/schools/:id	Comprehensive school profile with RTO license, dual-brake fleet, and verified reviews.
Courses & Pricing	/pricing	Transparent fee cards comparing 2-Wheeler (MCWG), 4-Wheeler (LMV), and Combo tracks.
Parivahan Mock Test	RTOQuiz.jsx	Interactive 15-question road sign and traffic simulator with immediate score results.
Launch Offers & Terms	/terms-and-offers	Full terms of introductory Rs.15 wallet bonuses, student safety rules, and cancellations.
Authentication Gateway	/login, /signup	Role-based authentication routing users directly to their designated dashboard.

2. Learner Hub Portal (/learner)

Operational User Journey Flow:
[Enrolled Student] | v +-----+ +-----+ +-----+
+-----+ | Real-Time Hub | ---> | Courses & Schools | ---> | Book Lesson | ---> | RTO Exam
Prep | ---> [Form 5 License] | (Monsoon Alerts) | | (Explore & Enroll)| | (Landmark Pickup)| | (Parivahan
Sarathi)| +-----+ +-----+ +-----+

Module / Feature	Route / Component	One-Line Technical & Operational Purpose
Learner Dashboard	/learner/dashboard	Central cockpit displaying current drive, monsoon rain delay alerts, and trainer remarks.
Courses & Schools Hub	/learner/courses	Native course discovery page allowing students to explore, compare, and enroll in new tracks.
Course Enrollment Modal	CourseEnrollModal.jsx	Package reservation modal prefilled with Parivahan LL data, batch timing, and wallet bonus.
My Practical Bookings	/learner/bookings	Complete schedule table to view upcoming drives, reschedule conflicts, or cancel slots.
Interactive Slot Booking	BookLessonModal.jsx	1-click modal to book next practical lesson with verified landmark pickup selection.
Curriculum Progress	/learner/progress	Visual step-by-step milestone stepper tracking student progress from basics to permanent DL.
RTO Mock Exam Portal	/learner/rto-mock-test	Complete exam portal with question palette, review flags, and pass/fail analysis.
In-App Wallet	/learner/wallet	Displays current balance, introductory credit deposit, and course payment histories.
Form 5 Certificate	/learner/certificates	Official motor training certificate under CMVR Rule 27 with QR code and PDF download.
Family Safety Shield	SafetyShieldCard.jsx	1-click WhatsApp emergency trigger sharing live location, car number, and trainer details.

3. Instructor Cockpit Portal (/instructor)

Operational User Journey Flow:
[Certified Instructor] | v +-----+ +-----+ +-----+
+-----+ | Today's Schedule | ---> | Transit Landmark | ---> | Live In-Cockpit | ---> |
Skill Checkpoint | ---> [Lesson Completed] | (Student Queue) | | (Pillar Pickup) | | (45-Min Timer) | |
(8-Track & Clutch) | +-----+ +-----+ +-----+

Module / Feature	Route / Component	One-Line Technical & Operational Purpose
Instructor Dashboard	/instructor/dashboard	High-level cockpit showing today's teaching load, pass rate metrics, and student alerts.
Today's Schedule	/instructor/schedule	Chronological queue of scheduled sessions with student phone dialer and landmark details.
In-Cockpit Live Console	StartLessonModal.jsx	Real-time 45-min drive console with safety drill checkboxes, remarks, and timer.
Weather Delay Broadcast	WeatherBroadcastModal	Dual-authority monsoon alert tool allowing instructors to broadcast delay warnings.

Student Skill Roster	/instructor/students	Digital skill evaluation matrix grading students on clutch biting, 8-track, and parking.
Slot Availability Matrix	/instructor/availability	Weekly recurring calendar manager to configure open and blocked training slots.
Trainer Chat Inbox	/instructor/messages	Direct messaging channel between instructors, learners, and school management.

4. School Owner Operations Portal (/owner)

Operational User Journey Flow:
[Driving School Owner] | v +-----+ +-----+ +-----+
+-----+ | Fleet Operations | ---> | Vehicle Dossier | ---> | Student KYC | ---> | Fuel / CNG Ledger | ---> [Revenue & P&L;] | (Dual-Control) | | (RC, RTO Fitness) | | (Sarathi LL Form)| | (KM / Liter Audit) | +-----+ +-----+ +-----+

Module / Feature	Route / Component	One-Line Technical & Operational Purpose
Owner Overview	/owner/dashboard	High-level overview showing active learners, fleet health, monthly revenue, and weather controls.
Fleet & Vehicle Registry	/owner/vehicles	Registry tracking dual-control status, insurance expiry, and fitness certification.
Vehicle Dossier Modal	VehicleDetailModal.jsx	Printable vehicle compliance dossier with RC, chassis, dual-brake cert, and ledger.
Student Admissions	/owner/students	Enrollment registry tracking batch assignments, fees paid, and Parivahan Form 2 documents.
Instructor Management	/owner/instructors	Staff manager overseeing instructor salaries, assigned training cars, and ratings.
Course Builder	/owner/courses	Interface to create and edit subsidized 2-wheeler, 4-wheeler, and combo packages.
Operational Bookings	/owner/bookings	Fleet-wide calendar showing all live slots across grounds and transit pickup gates.
Fuel & Financial Ledger	/owner/payments	Financial system featuring a vehicle-by-vehicle fuel/CNG expense and efficiency audit.
SaaS Platform Billing	/owner/subscription	Manages the school's DriveLearn platform subscription plan, limits, and renewal dates.
Student Reviews Desk	/owner/reviews	Review moderation center allowing owners to monitor and reply to student feedback.

5. Super Admin Governance Portal (/admin)

Operational User Journey Flow:
[RTO Transport Regulator / Super Admin] | v +-----+ +-----+
+-----+ +-----+ | Statewide State | ---> | School Inspection | ---> | Wallet Subsidies | ---> | Audit Trail | ---> [Dispute Closure] | (5 Districts) | | (License Verify) | | (Rs.15 Launch) | | (Immutable Logs) | +-----+ +-----+ +-----+

Module / Feature	Route / Component	One-Line Technical & Operational Purpose
State Command Center	/admin/dashboard	Statewide analytics tracking verified schools, registered learners, and platform revenue.
RTO Verification Desk	/admin/verification	Document inspection queue to verify and approve school licenses and dual-control certs.
Driving Schools List	/admin/schools	Full registry of driving schools across Maharashtra with operational status toggles.
Regional Coverage Hubs	/admin/locations	District management configuring Pune, Mumbai, Nagpur, Nashik, and Thane training zones.

Wallet & Subsidy Rules	/admin/offers	Central administration managing Rs.15 signup wallet credits and state launch discounts.
SaaS Subscriptions	/admin/subscriptions	Revenue engine tracking subscription tiers and payment status across all school owners.
Support & Disputes	/admin/support	Ticketing desk resolving student refund requests, trainer complaints, and RTO slot issues.
Compliance Audit Trail	/admin/audit-log	Immutable tamper-proof log recording all license approvals, payouts, and system edits.

6. Hyper-Local Engineering & Real-World Safety Innovations

- **Zero-Phone-Tag Transit Pickups (pickupLandmarks.js)**

Solves the primary real-world friction of students and trainers missing each other by enforcing verified metro pillar and ground gate landmarks (e.g. Garware College Metro Pillar 42, Warje 8-Track Gate 1).

- **Dual-Authority Monsoon Rain Delay System (weatherAlertState.js)**

Addresses seasonal Maharashtra flooding by empowering both instructors and owners to trigger a real-time rain delay broadcast, giving learners 1-click slot adjustments without penalty.

- **Digital In-Cockpit Console (StartLessonModal.jsx)**

Replaces paper punch cards with an active 45-minute cockpit timer, student Parivahan Sarathi LL verification badge, 5 mandatory drill checkboxes (helmet, clutch, 8-track), and instructor notes.

- **Family Reassurance Safety Shield (SafetyShieldCard.jsx)**

Tailored for female and first-time learners, providing a single-tap WhatsApp broadcast transmitting instructor name, vehicle registration number, and live training ground location to parents.

- **Parivahan Sarathi Regulatory Alignment**

Fully synchronized with Central Motor Vehicles Rules (CMVR 1989), supporting Form 2 learner license verification, simulated 8-track maneuvers, and official Form 5 completion certificates.